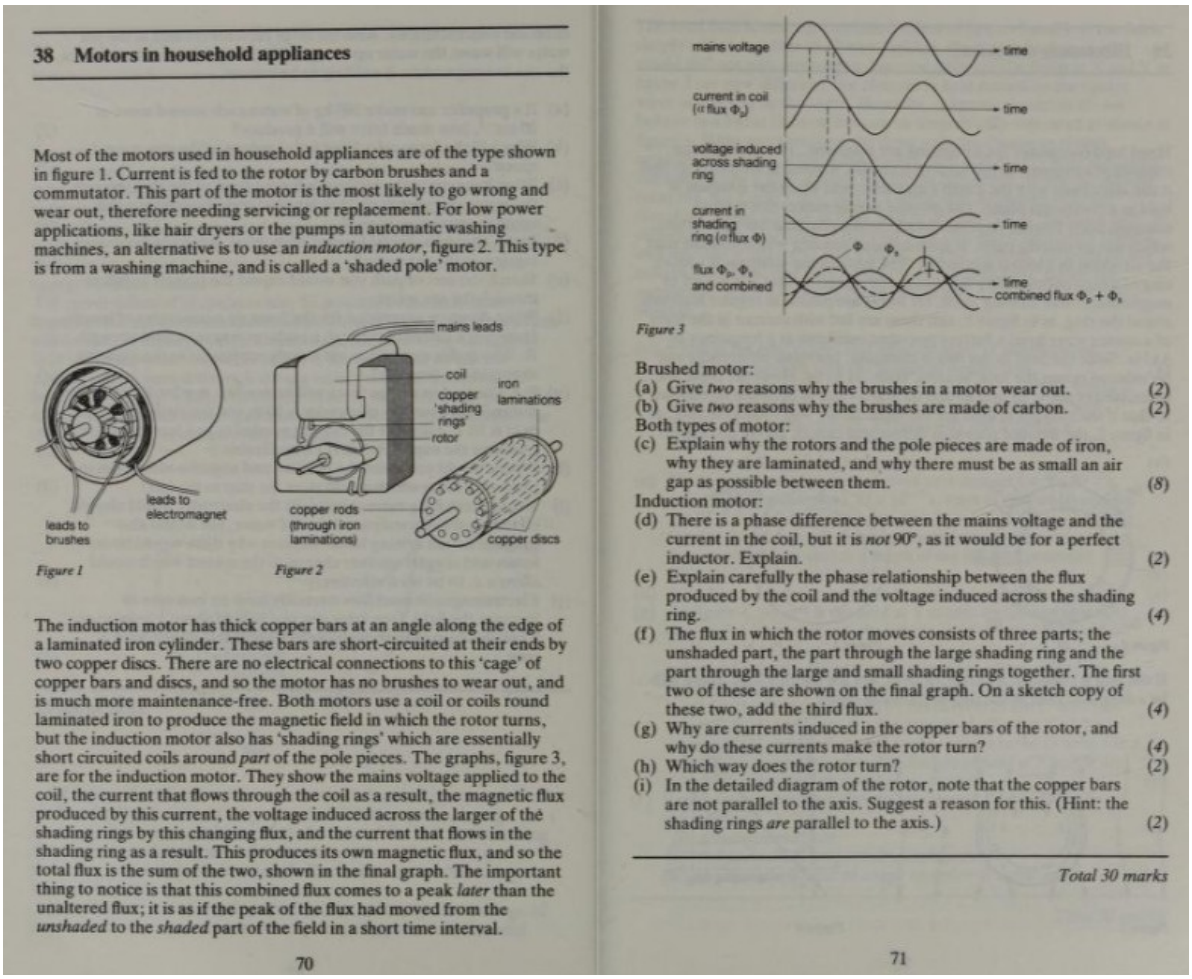
CIE 9702 Physics Paper 4 Style Question

Topic: Induction motors and phase relationships



38 Motors in household appliances

Most of the motors used in household appliances are of the type shown in figure 1. Current is fed to the rotor by carbon brushes and a commutator. This part of the motor is the most likely to go wrong and wear out, therefore needing servicing or replacement. For low power applications, like hair dryers or the pumps in automatic washing machines, an alternative is to use an *induction motor*, figure 2. This type is from a washing machine, and is called a 'shaded pole' motor.

Figure 1 *Figure 2*

The induction motor has thick copper bars at an angle along the edge of a laminated iron cylinder. These bars are short-circuited at their ends by two copper discs. There are no electrical connections to this 'cage' of copper bars and discs, and so the motor has no brushes to wear out, and is much more maintenance-free. Both motors use a coil or coils round laminated iron to produce the magnetic field in which the rotor turns, but the induction motor also has 'shading rings' which are essentially short circuited coils around *part* of the pole pieces. The graphs, figure 3, are for the induction motor. They show the mains voltage applied to the coil, the current that flows through the coil as a result, the magnetic flux produced by this current, the voltage induced across the larger of the shading rings by this changing flux, and the current that flows in the shading ring as a result. This produces its own magnetic flux, and so the total flux is the sum of the two, shown in the final graph. The important thing to notice is that this combined flux comes to a peak *later* than the unaltered flux; it is as if the peak of the flux had moved from the *unshaded* to the *shaded* part of the field in a short time interval.

70

Figure 3

Brushed motor:
(a) Give *two* reasons why the brushes in a motor wear out. (2)
(b) Give *two* reasons why the brushes are made of carbon. (2)

Both types of motor:
(c) Explain why the rotors and the pole pieces are made of iron, why they are laminated, and why there must be as small an air gap as possible between them. (8)

Induction motor:
(d) There is a phase difference between the mains voltage and the current in the coil, but it is *not* 90°, as it would be for a perfect inductor. Explain. (2)
(e) Explain carefully the phase relationship between the flux produced by the coil and the voltage induced across the shading ring. (4)
(f) The flux in which the rotor moves consists of three parts; the unshaded part, the part through the large shading ring and the part through the large and small shading rings together. The first two of these are shown on the final graph. On a sketch copy of these two, add the third flux. (4)
(g) Why are currents induced in the copper bars of the rotor, and why do these currents make the rotor turn? (4)
(h) Which way does the rotor turn? (2)
(i) In the detailed diagram of the rotor, note that the copper bars are not parallel to the axis. Suggest a reason for this. (Hint: the shading rings *are* parallel to the axis.) (2)

Total 30 marks

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Passage

Many household appliances use small AC induction motors because they are robust and require little maintenance. Unlike brushed motors, an induction motor has no brushes touching the rotor, so there is less frictional wear and reduced sparking. A common design is the **shaded-pole motor**. In this design, each pole has an unshaded region and a shaded region formed by a thick copper ring.

When alternating mains voltage is applied to the stator coil, the changing current creates a changing magnetic flux in the iron pole pieces. This changing flux induces an emf in the copper shading ring. The induced current in the ring produces its own magnetic flux, which is out of phase with the original stator flux. The combined effect is that the peak magnetic field appears first in the unshaded part and then in the shaded part, creating a weak sweeping field that can start rotor rotation.

A student investigates this process on a laboratory shaded-pole motor supplied by a 50 Hz source.

She measures the rms current in the stator coil and the induced current in the shading ring at different applied rms voltages. Her measurements are shown in Table 1.

Table 1: Experimental data for a shaded-pole motor at 50 Hz

V_{rms}/V	I_{coil}/A	I_{ring}/A	Rotor speed / rpm
120	0.32	0.68	860
160	0.43	0.89	1030
200	0.53	1.08	1190
240	0.64	1.29	1330

For one operating point, the student models the two flux components by

$$\phi_1 = \phi_0 \sin(\omega t), \qquad \phi_2 = 0.55\phi_0 \sin(\omega t - 55^\circ),$$

where ϕ_1 is the unshaded-pole flux and ϕ_2 is the shaded-pole flux.

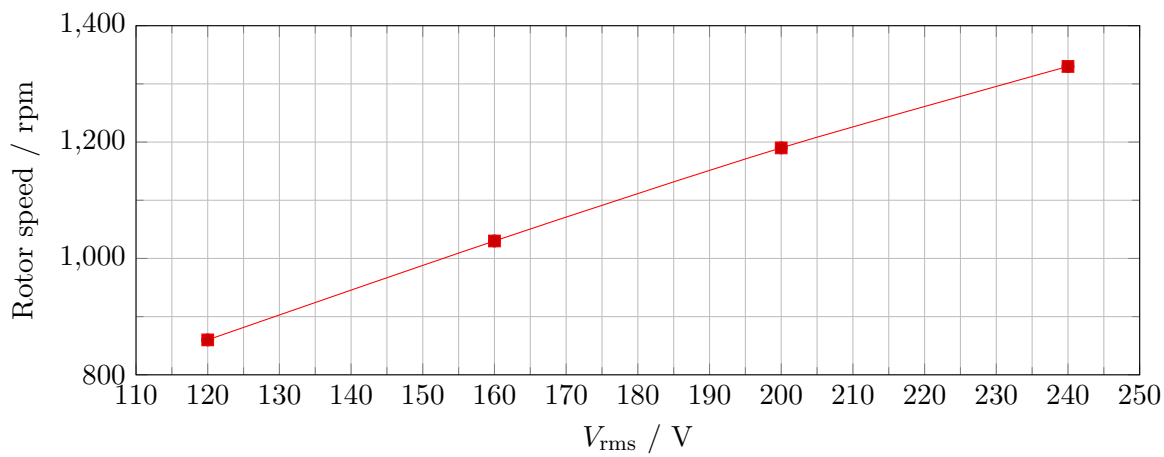

Figure 1: Rotor speed against applied rms voltage

Questions

1. Explain why a brushed motor is typically less suitable than an induction motor for a continuously operating household appliance such as a washing-machine pump. [3]

2. State two reasons why carbon is used for brushes in brushed motors. [2]

3. The rotor and stator cores are made of laminated soft iron rather than solid steel.

   (a) Explain why soft iron is used. [2]

   (b) Explain why lamination is required. [2]

4. Using the data in Table 1,

   (a) determine the gradient of the graph of rotor speed against V_{rms} between 120 V and 240 V, [2]

(b) estimate the rotor speed at 220 V. [2]

5. For the modeled fluxes,

   (a) determine the time delay Δt corresponding to a phase difference of 55° at 50 Hz, [2]

   (b) explain how this phase difference produces a starting torque on the rotor. [3]

6. At 240 V, the measured shading-ring current is 1.29 A. The resistance of the shading ring is 0.18 Ω.

   (a) calculate the power dissipated in the shading ring, [2]

   (b) suggest one design change to reduce this loss while maintaining motor start-up performance. [2]

7. **Open-ended question:**

   A manufacturer wants to redesign this shaded-pole motor for a desk fan that must be quieter and at least 10% more energy efficient, while still starting reliably under load.

   Propose and justify **two** engineering modifications. Your answer should refer to electromagnetic principles, likely effects on starting torque, efficiency, and any trade-offs in cost or complexity. [8]

Suggested answers (concise)

1. Brushed motors have mechanical contact between brushes and commutator, causing friction, wear, and sparking. This increases maintenance and can reduce reliability for long continuous operation. Induction motors avoid brush wear because torque is produced electromagnetically.

2. Carbon is a reasonable conductor but soft/lubricating, so commutator wear is reduced. It also tolerates heating and gives comparatively stable contact with lower risk of severe welding/sticking.

3. Soft iron magnetizes/demagnetizes easily (high permeability, low hysteresis loss), giving strong alternating flux. Lamination increases electrical resistance across the core and suppresses eddy currents, reducing heating losses.

4. (a) Gradient $\approx (1330 - 860)/(240 - 120) = 470/120 \approx 3.9\,\text{rpm V}^{-1}$.

   (b) Using near-linear trend: at 220 V, speed $\approx 860 + 3.9 \times 100 \approx 1250$ rpm (acceptable estimate around 1260 rpm).

5. (a) Period $T = 1/f = 0.020$ s. So $\Delta t = (55/360)T \approx 3.1 \times 10^{-3}$ s.

   (b) The shaded-ring flux lags the main pole flux, so flux density peaks shift spatially and temporally from unshaded to shaded side. Rotor bars experience changing flux linkage with a directional progression, producing induced currents and a net starting torque.

6. (a) $P = I^2R = (1.29)^2(0.18) \approx 0.30$ W.

   (b) Example: optimize ring cross-sectional area/material to reduce resistance and copper loss while preserving sufficient phase-shift current; or redesign pole geometry to improve useful rotating-field effect at lower ring current.

7. Open-ended responses should include two justified modifications. Creditworthy ideas include better laminated core material, optimized shading-ring geometry, improved bearing/friction design, auxiliary start winding with capacitor (if cost allows), and aerodynamic blade/fan loading optimization. Strong answers state benefits and trade-offs (cost, complexity, reduced start torque risk, noise changes).

Variant 2 (higher difficulty)

Passage

A second student studies a larger shaded-pole induction motor intended for continuous operation in a laboratory extraction fan. The supply frequency is fixed at 50 Hz, but the applied rms voltage is varied. At each setting, the student measures stator rms current I_{coil}, shading-ring current I_{ring}, shaft speed n, and real input power P_{in}.

For this motor, assume the synchronous speed is

$$n_s = 1500\,\text{rpm}.$$

Measured data are shown in Table 2.

Table 2: Extended data for Variant 2

V_{rms}/V	I_{coil}/A	I_{ring}/A	n/rpm	P_{in}/W
140	0.42	0.75	910	52
170	0.50	0.93	1040	67
200	0.59	1.11	1160	83
230	0.69	1.30	1270	101
250	0.76	1.43	1340	116

At 230 V, the output mechanical power is measured as 61 W. For the shading ring at this operating point, $R_{\text{ring}} = 0.16\,\Omega$.

The two flux components at one instant are modeled by

$$\phi_1 = \phi_0 \sin(\omega t), \qquad \phi_2 = 0.62\phi_0 \sin(\omega t - 63^\circ).$$

Questions (Variant 2)

1. Calculate the slip s at 200 V and at 250 V, where

$$s = \frac{n_s - n}{n_s}.$$

   State and explain the trend in slip as voltage increases. [4]

2. Determine the gradient of I_{ring} against V_{rms} between 170 V and 230 V. Use this gradient to estimate I_{ring} at 210 V. [3]

3. At 230 V,

   (a) calculate the motor efficiency η, [2]

   (b) calculate the shading-ring copper loss, [2]

   (c) determine this copper loss as a percentage of total input power. [2]

4. For a 50 Hz supply and a phase difference of 63° between ϕ_1 and ϕ_2,

(a) calculate the corresponding time delay, [2]

(b) explain why this delay is large enough to produce starting torque but still leads to energy dissipation in the ring. [3]

5. The speed measurement uncertainty is $\pm 10\,\text{rpm}$.

   (a) determine the percentage uncertainty in the speed reading at 910 rpm, [1]

   (b) use worst-case values to estimate the percentage uncertainty in the slip at 910 rpm. [3]

6. **Open-ended design question (higher difficulty):**

   The manufacturer now requires this motor to achieve **at least 15% reduction in total input power** at 230 V, while maintaining start-up torque and keeping the maximum steady-state temperature rise below current values.

   Propose and justify **two to three** design modifications. Your answer should integrate electromagnetic theory, loss mechanisms (copper, iron, mechanical), thermal implications, and practical engineering trade-offs (cost, manufacturability, reliability). [10]

Suggested answers (Variant 2, concise)

1. At 200 V: $s = (1500 - 1160)/1500 = 0.227 \approx 0.23$. At 250 V: $s = (1500 - 1340)/1500 = 0.107 \approx 0.11$. Slip decreases as voltage rises because rotor speed approaches synchronous speed (stronger effective rotating field and larger torque margin).

2. Gradient $\approx (1.30 - 0.93)/(230 - 170) = 0.37/60 = 6.17 \times 10^{-3}\,\text{A V}^{-1}$. From 170 V to 210 V (40 V increase): $\Delta I \approx 0.247\,\text{A}$, so $I_{\text{ring}}(210\,\text{V}) \approx 0.93 + 0.247 = 1.18\,\text{A}$.

3. (a) $\eta = P_{\text{out}}/P_{\text{in}} = 61/101 = 0.604 \approx 60\%$.

   (b) $P_{\text{ring}} = I^2 R = (1.30)^2(0.16) = 0.270\,\text{W}$.

   (c) Fraction $= 0.270/101 = 2.67 \times 10^{-3}$, so percentage $\approx 0.27\%$.

4. (a) $T = 1/50 = 0.020\,\text{s}$. $\Delta t = (63/360)T = 3.5 \times 10^{-3}\,\text{s}$.

   (b) Phase lag gives a time-shifted flux in shaded region, creating a sweeping field and start torque. But induced current in the resistive ring causes I^2R loss, so part of input power is dissipated as heat.

5. (a) Percentage uncertainty in n: $(10/910) \times 100\% \approx 1.10\%$.

   (b) Nominal $s = (1500 - 910)/1500 = 0.393$. Worst-case: $n = 920 \Rightarrow s_{\min} = 0.387$ and $n = 900 \Rightarrow s_{\max} = 0.400$. Absolute uncertainty $\approx \pm 0.0067$, so percentage uncertainty in slip $\approx (0.0067/0.393) \times 100\% \approx 1.7\%$.

6. Creditworthy proposals include improved magnetic steel (lower hysteresis/eddy loss), optimized lamination thickness, optimized shading-ring geometry/material, reduced bearing/friction losses, improved cooling path, or redesign to capacitor-assist start winding where acceptable. Strong responses quantify expected effects and discuss compromises in cost and manufacturing complexity.